Element E. Application of STEM Principles and Practices

STEM Principle for High-Priority Design Requirement #1
STEM Concept: Math
Describe how STEM concept applies directly to the specifics of your design:
In order to estimate how much materials we needed and to come up with dimensions, we had to multiply and calculate the area and diameter of our design. Since, multiplication, area, and diameter are parts of math, we have implemented math in our design and sketch.

STEM Principle for High-Priority Design Requirement #2
STEM Concept: Friction
Describe how STEM concept applies directly to the specifics of your design:
One of the constraints of the project was that the bookend had to resist sliding. This means that we had to come up with a way to add friction to our design. In order to accomplish this, we decided to add black rubber dots to the bottom of the bookend. The dots will create friction and make the bottom less flat. Since flat surfaces are slippery and easier to slide, the rubber provides friction for the sliding to stop or become less of a problem.

STEM Principle for High-Priority Design Requirement #3
STEM Concept: Mass and Force
Describe how STEM concept applies directly to the specifics of your design:
Another feature of the bookend is the ability to hold multiple books at a time. This means that the design has to hold a lot of mass. We have considered materials that are strong enough to withstand heavy things and switched our bonding methid to accommodate for this feature. Instead of using hot glue, we have decided to use wood glue since it is sturdier. Also, the orginal desing included popsicle sticks. However, we opted for wood to ensure the design was able to actually the keep the books. Lastly, the force of books, the bookends, and gravity were also taken into consideration. We understood that gravity will be a facor in keeping the books on the shelf. To prevent the shelves from collapsing we need to ensure that the bookend isn’t too heavy.

STEM Principle for Design Solution #1
STEM Concept: Mass
Describe how STEM concept applies directly to the specifics of your solution:
Ideally, the bookend should not be too heavy so it can be placed on shelves easily. We can calculate the mass of the design using mass = density x volume. Therefore, this a STEM concept that we will consider while building our prototype. Also, we have to consider the mass of the bookend because if it is too light, it might not hold enough books.

STEM Principle for Design Solution #2
STEM Concept: Volume and Density
Describe how STEM concept applies directly to the specifics of your solution:
We need to consider the volume of the bookend to ensure the product isn't too big or small. The design has to be big enough to fit multiple books in a row, but also small enough to fit on the book shelves in the library. Lastly, we can use volume to test what size is the most efficent for our goal of making functional and effective bookends. Density can be found in physics and chemistry. We need to understand and use this concept in order to calculate for the mass of the product. Also, density will help to ensure durability. We can remember that density and pressure are directly proportional. The more dense we make the bookend, the more pressure and weight it will have. Lastly, volume and density are indirectly proportional. This means the bigger the bookend becomes, the less dense it will be as long as the mass stays the same.

STEM Principle for Design Solution #3
STEM Concept: Durability
Describe how STEM concept applies directly to the specifics of your solution:
We need to ensure the bookends can not only contain multiple books at once, but last for long periods of time. The bookend should be able to last more than one school year. Engineering and how we build the product will determine the durability of the bookend. However, it is very important that our bookends can handle holding several books. We want our bookends to last and stay functional for a reasonable amount of time.